

Topic 4: Matrix Completion

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Learning goals. Understand matrix norms built from singular values; prove nuclear-norm convexity and its subgradient formula; derive singular-value thresholding (SVT); and use it to solve a matrix-completion objective by proximal gradient descent.

4.1 The problem and the notation

In matrix completion, only some entries of a matrix are observed. For example, a user-item table may contain many missing ratings. Without structural assumptions, the missing values are undetermined. An approximately low-rank model assumes that a small number of latent factors can explain much of the table.

We use M for the data and X for a candidate completed matrix. This keeps the data separate from the optimization variable throughout.

Symbol	Meaning
M, X, Y, G	Real $m \times n$ matrices: data, estimate, proximal input, and a subgradient or dual matrix, respectively
Ω	Set of observed index pairs (i, j)
$\mathcal{P}_\Omega, \mathcal{P}_{\Omega^c}$	Operators retaining observed and unobserved entries, respectively
r, q	$r = \text{rank}(X)$ and $q = \min\{m, n\}$
λ, t, τ	Regularization weight, step size, and SVT threshold; in proximal gradient, $\tau = t\lambda$
$\langle A, B \rangle_F$	Frobenius inner product $\text{tr}(A^\top B) = \sum_{i,j} A_{ij} B_{ij}$

Only M_{ij} with $(i, j) \in \Omega$ is used in the objective. Values stored at missing locations are placeholders, not observations.

4.1.1 Compact singular value decomposition

For a rank- r matrix X ,

$$X = U_r \Sigma_r V_r^\top = \sum_{i=1}^r \sigma_i u_i v_i^\top, \quad \Sigma_r = \text{diag}(\sigma_1, \dots, \sigma_r), \quad \sigma_1 \geq \dots \geq \sigma_r > 0,$$

where $U_r \in \mathbb{R}^{m \times r}$ and $V_r \in \mathbb{R}^{n \times r}$ have orthonormal columns. A thin SVD may instead retain q singular values, including zeros. For $X = 0$, the compact factors have no columns and their product is zero.

Road map. Sections 4.2–4.7 develop the nuclear norm and SVT. Sections 4.8–4.12 derive the completion algorithm and its objective convergence. A numerical example follows.

4.2 Three matrix norms and their singular-value meanings

Let $\sigma_1(X) \geq \dots \geq \sigma_q(X) \geq 0$ be the singular values, including zeros. Singular values are nonnegative even when X has negative entries or, for a symmetric matrix, negative eigenvalues.

Definition 4.1 (Spectral, Frobenius, and nuclear norms)

$$\|X\|_{\text{op}} := \max_{\|v\|_2=1} \|Xv\|_2 = \sigma_1(X), \quad (4.1)$$

$$\|X\|_F := \sqrt{\sum_{i,j} X_{ij}^2} = \sqrt{\sum_{i=1}^q \sigma_i(X)^2}, \quad (4.2)$$

$$\|X\|_* := \sum_{i=1}^q \sigma_i(X). \quad (4.3)$$

The nuclear norm is also called the trace norm or Schatten 1-norm. The spectral norm is also called the operator 2-norm.

Thus the three quantities are, respectively, the ℓ_∞ , ℓ_2 , and ℓ_1 norms of the vector of singular values. We use $\|\cdot\|_{\text{op}}$ to avoid confusing the spectral norm with the Frobenius norm.

4.2.1 Two examples

For $X = \text{diag}(3, -1)$, the singular values are 3 and 1, so

$$\|X\|_{\text{op}} = 3, \quad \|X\|_F = \sqrt{10}, \quad \|X\|_* = 4.$$

The trace is 2, which is different from the nuclear norm. For a symmetric positive semidefinite matrix, the eigenvalues are nonnegative, and its nuclear norm does equal its trace.

For $J = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, the singular values are 2, 0:

$$\|J\|_{\text{op}} = \|J\|_F = \|J\|_* = 2, \quad \sum_{i,j} |J_{ij}| = 4.$$

The nuclear norm is *not* the sum of absolute matrix entries. An entrywise ℓ_1 penalty encourages sparse entries; a nuclear-norm penalty encourages a sparse vector of singular values, hence low rank.

4.2.2 Useful inequalities

For a rank- r matrix with $r > 0$,

$$\|X\|_{\text{op}} \leq \|X\|_F \leq \|X\|_* \leq \sqrt{r} \|X\|_F.$$

The final inequality follows from Cauchy–Schwarz: $\sum_{i=1}^r \sigma_i \leq \sqrt{r} (\sum_{i=1}^r \sigma_i^2)^{1/2}$. All norms vanish at the zero matrix.

Rank counts; the nuclear norm sums. $\text{rank}(X) = \#\{i : \sigma_i(X) > 0\}$ counts nonzero singular values. The convex surrogate $\|X\|_* = \sum_i \sigma_i(X)$ also depends on their magnitudes. Penalizing the nuclear norm promotes low rank but is not the same as imposing a chosen rank.

4.3 Duality proves the triangle inequality and convexity

Theorem 4.2 (Nuclear–spectral duality) For every $X \in \mathbb{R}^{m \times n}$,

$$\|X\|_* = \max_{\|G\|_{\text{op}} \leq 1} \langle G, X \rangle_F. \quad (4.4)$$

Proof. Write the compact SVD $X = \sum_{i=1}^r \sigma_i u_i v_i^\top$. If $\|G\|_{\text{op}} \leq 1$, then

$$\langle G, X \rangle_F = \sum_{i=1}^r \sigma_i u_i^\top G v_i \leq \sum_{i=1}^r \sigma_i |u_i^\top G v_i| \leq \sum_{i=1}^r \sigma_i = \|X\|_*.$$

The last inequality uses $|u_i^\top G v_i| \leq \|u_i\|_2 \|G\|_{\text{op}} \|v_i\|_2 \leq 1$. For $X \neq 0$, choose $G = U_r V_r^\top$. This matrix has spectral norm 1 and $u_i^\top G v_i = 1$, so the upper bound is attained. For $X = 0$, both sides are zero. ■

Rescaling gives the matrix Hölder inequality

$$|\langle G, X \rangle_F| \leq \|G\|_{\text{op}} \|X\|_*. \quad (4.5)$$

Conversely, $\|G\|_{\text{op}} = \max_{\|X\|_* \leq 1} \langle G, X \rangle_F$: the upper bound follows from (4.5), and a leading singular-vector outer product of G attains it.

4.3.1 Why the nuclear norm is a norm

Singular values give nonnegativity and $\|X\|_* = 0 \iff X = 0$. For every real scalar a ,

$$\|aX\|_* = |a| \|X\|_*,$$

because the singular values are multiplied by $|a|$, not by a . Finally, duality proves the triangle inequality without assuming it:

$$\begin{aligned} \|A + B\|_* &= \max_{\|G\|_{\text{op}} \leq 1} (\langle G, A \rangle_F + \langle G, B \rangle_F) \\ &\leq \max_{\|G\|_{\text{op}} \leq 1} \langle G, A \rangle_F + \max_{\|G\|_{\text{op}} \leq 1} \langle G, B \rangle_F = \|A\|_* + \|B\|_*. \end{aligned}$$

4.3.2 Convexity

For $0 \leq \alpha \leq 1$, the triangle inequality and homogeneity give

$$\|\alpha A + (1 - \alpha)B\|_* \leq \alpha \|A\|_* + (1 - \alpha) \|B\|_*.$$

Thus the nuclear norm is convex. Another interpretation of (4.4) is that it is the pointwise maximum of linear functions of X , which is also a direct reason for convexity.

4.4 Subgradients: replacing a derivative at a corner

Definition 4.3 (Matrix subgradient) For a convex function $h : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$, a matrix G is a subgradient at X if

$$h(Z) \geq h(X) + \langle G, Z - X \rangle_F \quad \text{for every } Z. \quad (4.6)$$

The set of all such matrices is the subdifferential, denoted $\partial h(X)$.

For a differentiable convex function, the subdifferential contains only the gradient. At a nondifferentiable point, it can contain many matrices. For example, $\partial|x| = \{1\}$ for $x > 0$, $\{-1\}$ for $x < 0$, and $[-1, 1]$ at $x = 0$. Convex optimality is expressed by

$$X^* \text{ minimizes } h \iff 0 \in \partial h(X^*).$$

Indeed, setting $G = 0$ in (4.6) is exactly global minimality.

Lemma 4.4 (Subgradients of the nuclear norm)

$$G \in \partial \|X\|_* \iff \|G\|_{\text{op}} \leq 1, \quad \langle G, X \rangle_F = \|X\|_*. \quad (4.7)$$

Proof. Sufficiency. By (4.5), for every Z ,

$$\|Z\|_* \geq \langle G, Z \rangle_F = \langle G, X \rangle_F + \langle G, Z - X \rangle_F = \|X\|_* + \langle G, Z - X \rangle_F.$$

Necessity. If G is a subgradient, apply (4.6) first with $Z = 0$ and then with $Z = 2X$:

$$0 \geq \|X\|_* - \langle G, X \rangle_F, \quad 2\|X\|_* \geq \|X\|_* + \langle G, X \rangle_F.$$

Together these imply $\langle G, X \rangle_F = \|X\|_*$. The subgradient inequality now reduces to $\|Z\|_* \geq \langle G, Z \rangle_F$ for every Z . Maximizing over $\|Z\|_* \leq 1$ and using duality gives $\|G\|_{\text{op}} \leq 1$. ■

Thus the subgradients are exactly the maximizers in the dual representation of the nuclear norm. At the zero matrix,

$$\partial \|0\|_* = \{G : \|G\|_{\text{op}} \leq 1\}. \quad (4.8)$$

A set, not an ordinary derivative. The notation $\partial \|X\|_*$ denotes a set of matrices. Statements such as $G \in \partial \|X\|_*$ and $0 \in X - Y + \tau \partial \|X\|_*$ are set-membership statements. This distinction matters when singular values vanish.

4.5 The full nuclear-norm subgradient formula

Theorem 4.5 *Let $X = U_r \Sigma_r V_r^\top$ be a compact SVD using only positive singular values. Then*

$$\partial \|X\|_* = \{U_r V_r^\top + W : U_r^\top W = 0, \quad W V_r = 0, \quad \|W\|_{\text{op}} \leq 1\}. \quad (4.9)$$

The term $U_r V_r^\top$ acts on the existing singular-vector directions. The matrix W acts between their orthogonal complements. The two orthogonality conditions are essential.

Proof. Every displayed matrix is a subgradient. Set $G = U_r V_r^\top + W$ with the stated conditions. Since $W V_r = 0$, we have $\langle G, X \rangle_F = \sum_{i=1}^r \sigma_i = \|X\|_*$. For any vector z , decompose it into its component in the column space of V_r and its orthogonal component:

$$Gz = U_r V_r^\top z + W(I_n - V_r V_r^\top)z.$$

These two output vectors are orthogonal because $U_r^\top W = 0$. Therefore

$$\begin{aligned} \|Gz\|_2^2 &= \|V_r^\top z\|_2^2 + \|W(I_n - V_r V_r^\top)z\|_2^2 \\ &\leq \|V_r V_r^\top z\|_2^2 + \|(I_n - V_r V_r^\top)z\|_2^2 = \|z\|_2^2. \end{aligned}$$

Thus $\|G\|_{\text{op}} \leq 1$, and Lemma 4.4 applies.

Every subgradient has this form. For a subgradient G , that lemma gives $\|G\|_{\text{op}} \leq 1$ and

$$\sum_{i=1}^r \sigma_i u_i^\top G v_i = \sum_{i=1}^r \sigma_i.$$

Each $u_i^\top G v_i \leq 1$ and every $\sigma_i > 0$, so $u_i^\top G v_i = 1$ for all i . Equality in Cauchy–Schwarz, together with $\|G v_i\|_2 \leq 1$, forces $G v_i = u_i$. Applying the same argument to $G^\top$ gives $G^\top u_i = v_i$. Hence $G V_r = U_r$ and $U_r^\top G = V_r^\top$. Define

$$W = G - U_r V_r^\top = (I_m - U_r U_r^\top) G (I_n - V_r V_r^\top).$$

Then $W V_r = 0$ and $U_r^\top W = 0$. Orthogonal projections do not increase the spectral norm, so $\|W\|_{\text{op}} \leq \|G\|_{\text{op}} \leq 1$. ■

Example. At $X = \text{diag}(3, 0)$,

$$\partial \|X\|_* = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & w \end{pmatrix} : |w| \leq 1 \right\}.$$

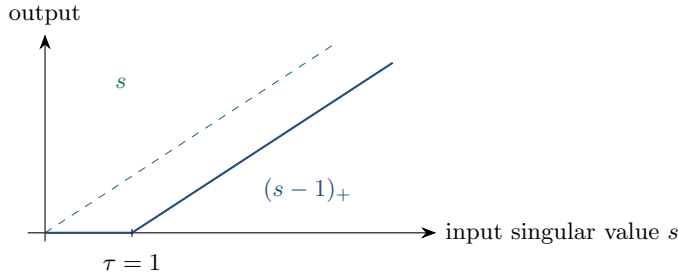
At $X = 0$, the formula reduces to (4.8). Use the compact SVD of the matrix *at which the subgradient is taken*; including zero-singular-value directions in U_r, V_r would impose incorrect restrictions on W .

4.6 Singular-value thresholding and the proximal operator

For a scalar s , define its positive part by $(s)_+ = \max\{s, 0\}$. Given a thin SVD $Y = U \text{diag}(\sigma_1, \dots, \sigma_q) V^\top$ and a threshold $\tau \geq 0$, define

$$\mathcal{S}_\tau(Y) = U \text{diag}((\sigma_1 - \tau)_+, \dots, (\sigma_q - \tau)_+) V^\top. \quad (4.10)$$

This is *singular-value soft thresholding*, or singular-value shrinkage. It preserves the singular-vector directions, subtracts τ from larger singular values, and replaces singular values at or below τ by zero. The resulting matrix is independent of the choice of SVD factors.



4.6.1 A concrete example

For $Y = \text{diag}(3, -1)$ and $\tau = 2$, an SVD is $U = I_2$, $\Sigma = \text{diag}(3, 1)$, $V^\top = \text{diag}(1, -1)$. Thus

$$\mathcal{S}_2(Y) = I_2 \text{diag}(1, 0) \text{diag}(1, -1) = \text{diag}(1, 0).$$

For $\tau = 1/2$, the result is $\text{diag}(2.5, -0.5)$. The negative sign belongs to the singular vectors, not to a negative singular value. In general,

$$\text{rank}(\mathcal{S}_\tau(Y)) = \#\{i : \sigma_i(Y) > \tau\}, \quad \mathcal{S}_\tau(Y) = 0 \iff \|Y\|_{\text{op}} \leq \tau.$$

Definition 4.6 (Proximal operator) For a convex penalty h and parameter $t > 0$,

$$\text{prox}_{th}(Y) := \arg \min_Z \left\{ \frac{1}{2} \|Z - Y\|_F^2 + t h(Z) \right\}.$$

For the finite convex penalties used here, the quadratic term ensures a unique minimizer.

The key result, proved next, is

$$\text{prox}_{\tau\|\cdot\|_*}(Y) = \mathcal{S}_\tau(Y).$$

This shrinks singular values, rather than applying scalar thresholding to each matrix entry or retaining the top r singular values unchanged.

4.7 Why SVT solves the nuclear-norm proximal problem

Theorem 4.7 (SVT is the nuclear-norm proximal map) For $\tau \geq 0$ and any $Y \in \mathbb{R}^{m \times n}$,

$$\mathcal{S}_\tau(Y) = \arg \min_Z \left\{ \frac{1}{2} \|Z - Y\|_F^2 + \tau \|Z\|_* \right\}. \quad (4.11)$$

The minimizer is unique.

Proof. The objective is continuous, coercive, and 1-strongly convex in the Frobenius norm, so it has exactly one minimizer. When $\tau = 0$, that minimizer is $Y = \mathcal{S}_0(Y)$. Suppose now that $\tau > 0$. Convex optimality gives

$$0 \in Z - Y + \tau \partial \|Z\|_* \iff \frac{Y - Z}{\tau} \in \partial \|Z\|_*. \quad (4.12)$$

We verify this condition at $Z = \mathcal{S}_\tau(Y)$.

Split the singular directions. Partition the compact SVD of Y into singular values strictly greater than τ and those at most τ :

$$Y = U_+ \Sigma_+ V_+^\top + U_- \Sigma_- V_-^\top, \quad \sigma_i(\Sigma_+) > \tau, \quad 0 < \sigma_i(\Sigma_-) \leq \tau.$$

Either part can be empty. Zero singular values need not be included. The thresholded matrix is

$$Z_\tau = \mathcal{S}_\tau(Y) = U_+ (\Sigma_+ - \tau I) V_+^\top.$$

Its *positive* singular directions are precisely U_+, V_+ .

Construct the required subgradient. Subtracting gives

$$\frac{Y - Z_\tau}{\tau} = U_+ V_+^\top + \underbrace{\frac{1}{\tau} U_- \Sigma_- V_-^\top}_W.$$

The SVD orthogonality relations imply

$$U_+^\top W = 0, \quad W V_+ = 0, \quad \|W\|_{\text{op}} \leq 1,$$

because every singular value in Σ_- is at most τ . Theorem 4.5, applied at Z_τ , therefore yields

$$\frac{Y - Z_\tau}{\tau} \in \partial \|Z_\tau\|_*.$$

This proves optimality by (4.12); strong convexity proves uniqueness. ■

What happens at the threshold? A singular value equal to τ is removed from Z_τ and contributes a singular value equal to 1 to W , which is allowed. If all singular values are removed, use $\partial \|0\|_* = \{G : \|G\|_{\text{op}} \leq 1\}$. Thus the proof covers ties and a zero output.

4.8 The observation operator and the completion objective

Define the observation operator entry by entry:

$$[\mathcal{P}_\Omega(X)]_{ij} = \begin{cases} X_{ij}, & (i, j) \in \Omega, \\ 0, & (i, j) \notin \Omega. \end{cases} \quad \mathcal{P}_{\Omega^c}(X) := X - \mathcal{P}_\Omega(X). \quad (4.13)$$

It is linear, self-adjoint in the Frobenius inner product, and idempotent:

$$\langle \mathcal{P}_\Omega(A), B \rangle_F = \langle A, \mathcal{P}_\Omega(B) \rangle_F, \quad \mathcal{P}_\Omega(\mathcal{P}_\Omega(A)) = \mathcal{P}_\Omega(A).$$

Also, $\mathcal{P}_\Omega(A)$ and $\mathcal{P}_{\Omega^c}(A)$ have disjoint supports, so

$$\|A\|_F^2 = \|\mathcal{P}_\Omega(A)\|_F^2 + \|\mathcal{P}_{\Omega^c}(A)\|_F^2, \quad \|\mathcal{P}_\Omega(A)\|_F \leq \|A\|_F.$$

4.8.1 A convex, penalized matrix-completion model

For a fixed $\lambda > 0$, solve

$$\min_X F(X) := \underbrace{\frac{1}{2} \|\mathcal{P}_\Omega(X - M)\|_F^2}_{f(X)} + \underbrace{\lambda \|X\|_*}_{g(X)}. \quad (4.14)$$

Equivalently, $f(X) = \frac{1}{2} \sum_{(i,j) \in \Omega} (X_{ij} - M_{ij})^2$. Only observed entries contribute to the loss; the nuclear norm couples all entries through the singular values. Both terms are convex, while g can be nondifferentiable.

4.8.2 Gradient and its Lipschitz constant

Differentiating the entrywise sum, or using self-adjointness, gives

$$\nabla f(X) = \mathcal{P}_\Omega(X - M). \quad (4.15)$$

For any X_1, X_2 ,

$$\|\nabla f(X_1) - \nabla f(X_2)\|_F = \|\mathcal{P}_\Omega(X_1 - X_2)\|_F \leq \|X_1 - X_2\|_F.$$

Thus $L = 1$ is a valid global gradient-Lipschitz constant. If Ω is nonempty, it is the smallest such constant; if Ω is empty, the gradient is zero and $L = 1$ is still a valid upper bound.

Existence and curvature. Since $F(X) \geq \lambda \|X\|_* \geq \lambda \|X\|_F$, the objective tends to infinity as $\|X\|_F \rightarrow \infty$. Continuity therefore ensures a minimizer. Missing entries create directions of zero curvature in the data-fit term, so strong convexity and uniqueness should not be assumed.

Missing is different from observed zero. A zero can be a genuine observed value. Store Ω separately from the numerical data. Setting missing values to zero in storage is harmless only when the observation mask is still used in the loss and gradient.

4.9 From a smooth upper bound to proximal gradient

Consider $F = f + g$, with f convex and L -smooth in the Frobenius norm, and g convex. We retain the nonsmooth term g exactly and approximate only the smooth part.

Lemma 4.8 (Descent lemma) *For an L -smooth function,*

$$f(Z) \leq f(X) + \langle \nabla f(X), Z - X \rangle_F + \frac{L}{2} \|Z - X\|_F^2. \quad (4.16)$$

Proof. Set $D = Z - X$. Integrating along the segment gives

$$\begin{aligned} f(Z) - f(X) - \langle \nabla f(X), D \rangle_F &= \int_0^1 \langle \nabla f(X + sD) - \nabla f(X), D \rangle_F ds \\ &\leq \int_0^1 Ls \|D\|_F^2 ds = \frac{L}{2} \|D\|_F^2. \end{aligned}$$

The inequality uses Frobenius Cauchy–Schwarz and the Lipschitz gradient bound. ■

4.9.1 Build and minimize an upper model

For a step size $0 < t \leq 1/L$, define

$$Q_t(Z; X) = f(X) + \langle \nabla f(X), Z - X \rangle_F + \frac{1}{2t} \|Z - X\|_F^2 + g(Z). \quad (4.17)$$

The descent lemma gives $F(Z) \leq Q_t(Z; X)$ and $Q_t(X; X) = F(X)$. Set $X^+ = \arg \min_Z Q_t(Z; X)$. Then

$$F(X^+) \leq Q_t(X^+; X) \leq Q_t(X; X) = F(X),$$

which proves nonincrease of the objective.

4.9.2 Complete the square carefully

The terms depending on Z in the model can be written as

$$\frac{1}{2t} \|Z - (X - t\nabla f(X))\|_F^2 + g(Z) \quad \text{plus a constant independent of } Z.$$

Multiplying by t does not change the minimizer, giving

$$\boxed{X^+ = \text{prox}_{tg}(X - t\nabla f(X)).} \quad (4.18)$$

For $g(Z) = \lambda \|Z\|_*$, Theorem 4.7 yields

$$\boxed{X^+ = \mathcal{S}_{t\lambda}(X - t\nabla f(X)).} \quad (4.19)$$

Track both uses of the step size. The gradient step uses t , and the singular-value threshold is $t\lambda$. At $t = 1/L$, the threshold is λ/L . It becomes λ only when $t = 1$.

4.10 The matrix-completion update: fill in, then shrink

Substitute $\nabla f(X) = \mathcal{P}_\Omega(X - M)$ into (4.19):

$$X^{(k+1)} = \mathcal{S}_{t\lambda} \left(X^{(k)} - t\mathcal{P}_\Omega(X^{(k)} - M) \right), \quad 0 < t \leq 1. \quad (4.20)$$

The choice $t = 1$ is valid because $L = 1$ for (4.14). Using $X - \mathcal{P}_\Omega(X) = \mathcal{P}_{\Omega^c}(X)$, it simplifies to

$$\boxed{X^{(k+1)} = \mathcal{S}_\lambda \left(\mathcal{P}_\Omega(M) + \mathcal{P}_{\Omega^c}(X^{(k)}) \right)}. \quad (4.21)$$

The matrix supplied to SVT has observed entries from the data and missing entries from the current estimate. This iteration is commonly called *Soft-Impute*; see the further reading at the end.

4.10.1 An explicit algorithm

Inputs: observed values $\{M_{ij} : (i, j) \in \Omega\}$, $\lambda > 0$, and a stopping tolerance. Initialize $X^{(0)} = 0$.

1. Form the filled matrix

$$Y^{(k)} = \mathcal{P}_\Omega(M) + \mathcal{P}_{\Omega^c}(X^{(k)}), \quad Y_{ij}^{(k)} = \begin{cases} M_{ij}, & (i, j) \in \Omega, \\ X_{ij}^{(k)}, & (i, j) \notin \Omega. \end{cases}$$

2. Compute a thin SVD $Y^{(k)} = U \text{diag}(\sigma_i) V^\top$.
3. Set $X^{(k+1)} = U \text{diag}((\sigma_i - \lambda)_+) V^\top$.
4. Check the step residual or a prescribed iteration limit. If continuing, increment k and repeat.

Output: the estimated matrix, including its entries on Ω^c .

4.10.2 What the two stages do

The filling stage restores observed entries *before* the SVD. The SVT stage can change every entry, including observed ones, because it fits a global low-rank structure. Exact agreement with observations is not a constraint in the penalized objective (4.14). One initial zero-fill followed by one SVD is generally not the solution: the missing entries must be updated repeatedly.

Scaling changes the step and threshold. If the loss is instead $f(X) = \|\mathcal{P}_\Omega(X - M)\|_F^2 / (2|\Omega|)$ with $|\Omega| > 0$, then $L = 1/|\Omega|$. Choosing $t = |\Omega|$ still gives the same filled matrix, but the threshold is now $|\Omega|\lambda$ when the penalty remains $\lambda \|X\|_*$. Specify the loss normalization when reporting λ .

4.11 A one-step inequality and a precise descent guarantee

The upper-model argument proves nonincrease. Convexity also gives a quantitative decrease and the key inequality needed for a convergence rate. This section applies to $F = f + g$ with convex L -smooth f , convex g , and $0 < t \leq 1/L$.

Lemma 4.9 (Comparison with any candidate matrix) *Let $X^+ = \arg \min_Z Q_t(Z; X)$. For every matrix Z ,*

$$\boxed{F(X^+) - F(Z) \leq \frac{1}{2t} \left(\|X - Z\|_F^2 - \|X^+ - Z\|_F^2 \right)}. \quad (4.22)$$

Proof. As a function of Z , $Q_t(Z; X)$ is $1/t$ -strongly convex. At its minimizer X^+ , strong convexity gives

$$Q_t(Z; X) \geq Q_t(X^+; X) + \frac{1}{2t} \|Z - X^+\|_F^2. \quad (4.23)$$

The linear term in this strong-convexity lower bound vanishes because zero is a subgradient at the minimizer X^+ . Also, the convex first-order inequality for f is

$$f(X) + \langle \nabla f(X), Z - X \rangle_F \leq f(Z).$$

Substitute this into the definition of Q_t to obtain

$$Q_t(Z; X) \leq F(Z) + \frac{1}{2t} \|Z - X\|_F^2.$$

Combine this bound with (4.23) and the majorization $F(X^+) \leq Q_t(X^+; X)$:

$$\begin{aligned} F(X^+) &\leq Q_t(X^+; X) \\ &\leq Q_t(Z; X) - \frac{1}{2t} \|Z - X^+\|_F^2 \\ &\leq F(Z) + \frac{1}{2t} \|Z - X\|_F^2 - \frac{1}{2t} \|Z - X^+\|_F^2. \end{aligned}$$

Rearranging proves the claim. ■

4.11.1 Set the comparison point to the current iterate

Taking $Z = X$ yields

$$\boxed{F(X^+) \leq F(X) - \frac{1}{2t} \|X^+ - X\|_F^2}. \quad (4.24)$$

The decrease is strict unless the update leaves X unchanged. For matrix completion at $t = 1$, the guaranteed decrease is at least $\frac{1}{2} \|X^{(k+1)} - X^{(k)}\|_F^2$.

4.11.2 A fixed point is an optimum

The model's optimality condition is

$$0 \in \nabla f(X) + \frac{X^+ - X}{t} + \partial g(X^+).$$

If $X^+ = X$, this becomes $0 \in \nabla f(X) + \partial g(X)$, the global optimality condition for convex F . Conversely, an optimizer satisfies the fixed-point equation because the proximal subproblem has a unique solution. Thus a zero step characterizes optimality, not merely a stalled update.

4.12 Sublinear objective convergence and stopping checks

Theorem 4.10 (Proximal-gradient objective rate) *Let f be convex and L -smooth, let g be convex, and suppose $F = f + g$ has a minimizer X^* . With a fixed step $0 < t \leq 1/L$ and exact proximal updates, for $k \geq 1$,*

$$F(X^{(k)}) - F(X^*) \leq \frac{\|X^{(0)} - X^*\|_F^2}{2tk}. \quad (4.25)$$

Proof. Write $\Delta_j = F(X^{(j)}) - F(X^*)$ and $D_j = \|X^{(j)} - X^*\|_F$. Apply (4.22) with comparison point $Z = X^*$ at each iteration:

$$\Delta_{j+1} \leq \frac{1}{2t}(D_j^2 - D_{j+1}^2).$$

Summing for $j = 0, \dots, k-1$ cancels all intermediate squared distances:

$$\sum_{j=0}^{k-1} \Delta_{j+1} \leq \frac{D_0^2 - D_k^2}{2t} \leq \frac{D_0^2}{2t}.$$

The gaps are nonnegative and nonincreasing by (4.24), so every term in this sum is at least Δ_k . Hence $k\Delta_k \leq D_0^2/(2t)$, proving the result. $\blacksquare$

For (4.14), $\lambda > 0$ ensures existence of X^* and $t = 1$ is valid. The bound is then $F(X^{(k)}) - F(X^*) \leq \|X^{(0)} - X^*\|_F^2/(2k)$. This is a last-iterate objective bound. It does not assert a unique completion, a linear rate, or recovery of an unknown true matrix.

4.12.1 A computable residual

The proximal-gradient mapping is

$$\mathcal{G}_t(X) := \frac{1}{t} [X - \text{prox}_{tg}(X - t\nabla f(X))] = \frac{X - X^+}{t}.$$

It is zero exactly at an optimizer. For the $t = 1$ algorithm, a practical relative step check is

$$\frac{\|X^{(k+1)} - X^{(k)}\|_F}{\max\{1, \|X^{(k)}\|_F\}} \leq \epsilon.$$

Use a maximum iteration count as well. This numerical tolerance is not, by itself, a certificate that the objective gap is at most ϵ .

4.12.2 Two useful exact checks

If every entry is observed, $X^{(k)} - \nabla f(X^{(k)}) = M$, and one step gives the exact minimizer $\mathcal{S}_\lambda(M)$. For an arbitrary mask, $X = 0$ is optimal if and only if

$$\lambda \geq \|\mathcal{P}_\Omega(M)\|_{\text{op}}. \quad (4.26)$$

Indeed, optimality at zero is $\mathcal{P}_\Omega(M) \in \lambda\partial\|0\|_*$, and (4.8) turns this into the spectral-norm inequality.

4.13 Worked example: one missing entry

Suppose the observed entries are

$$\text{data} = \begin{pmatrix} 1 & 1 \\ 1 & ? \end{pmatrix}, \quad \Omega = \{(1,1), (1,2), (2,1)\}, \quad \lambda = \frac{1}{2}.$$

Initialize $X^{(0)} = 0$ and use $t = 1$. The question mark denotes a missing entry; it is not an observed zero.

4.13.1 First fill and threshold

The first filled matrix and one thin SVD (rounded) are

$$\begin{aligned} Y^{(0)} &= \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = U\Sigma V^\top, \\ U &\approx \begin{pmatrix} 0.850651 & 0.525731 \\ 0.525731 & -0.850651 \end{pmatrix}, \quad V^\top \approx \begin{pmatrix} 0.850651 & 0.525731 \\ -0.525731 & 0.850651 \end{pmatrix}, \\ \Sigma &= \text{diag}\left(\frac{1+\sqrt{5}}{2}, \frac{\sqrt{5}-1}{2}\right) \approx \text{diag}(1.618034, 0.618034). \end{aligned}$$

Subtracting 0.5 from each singular value and reconstructing gives

$$X^{(1)} = \mathcal{S}_{0.5}(Y^{(0)}) \approx \begin{pmatrix} 0.776393 & 0.552786 \\ 0.552786 & 0.223607 \end{pmatrix}.$$

The missing entry is now nonzero, even though the gradient at that location was zero. The SVD step couples the entries through shared singular vectors.

4.13.2 Second fill and subsequent iterations

Restore the observed values while retaining the current missing estimate:

$$Y^{(1)} = \mathcal{P}_\Omega(M) + \mathcal{P}_{\Omega^c}(X^{(1)}) \approx \begin{pmatrix} 1 & 1 \\ 1 & 0.223607 \end{pmatrix}.$$

Another SVT step gives $X^{(2)} \approx \begin{pmatrix} 0.806583 & 0.552113 \\ 0.552113 & 0.377926 \end{pmatrix}$. The table uses the objective (4.14), including its penalty.

k	Missing estimate $X_{22}^{(k)}$	$F(X^{(k)})$	$\text{rank}(X^{(k)})$
0	0.000000	1.500000	0
1	0.223607	0.843034	2
2	0.377926	0.811562	1
3	0.434530	0.805694	1
4	0.457310	0.804795	1
5	0.466795	0.804642	1

The objective decreases, but the rank need not decrease at every iteration: it first rises from 0 to 2. The observed entries are generally shrunk away from 1 because the objective balances data fit and the nuclear-norm penalty. The missing estimate is a regularized prediction, not a guaranteed exact value of an underlying matrix.

4.14 Implementation checks and review questions

4.14.1 Checks that prevent common mistakes

1. **Specify the norm.** The subgradient condition uses $\|W\|_{\text{op}} \leq 1$; the data-fit loss uses $\|\cdot\|_F^2$; the regularizer uses $\|\cdot\|_*$.
2. **Keep the mask.** Missing entries contribute no residual. An observed zero must still be included in Ω .
3. **Threshold singular values.** Use $(\sigma_i - t\lambda)_+$, preserving the singular-vector pairing. Entrywise soft thresholding solves a different proximal problem.
4. **Use the matching objective.** Monitor both the observed-entry loss and $\lambda \|X\|_*$. These components need not decrease separately.
5. **Respect the proximal solve.** The stated proofs assume exact SVT. A partial SVD can reproduce it when all singular values above the threshold are retained and all omitted values are known to be at most the threshold. Approximate SVDs require separate error control.
6. **Distinguish two formulations.** The penalized problem (4.14) differs from $\min_X \|X\|_*$ subject to $\mathcal{P}_\Omega(X) = \mathcal{P}_\Omega(M)$. The latter enforces exact agreement on observations.

4.14.2 Questions for discussion

1. Compute the spectral, Frobenius, and nuclear norms of $\text{diag}(4, -2, 0)$. Why does the sign of an eigenvalue not change a singular value's nonnegativity?
2. Prove $\|aX\|_* = |a| \|X\|_*$ for negative as well as positive a . Then use duality to prove the triangle inequality.
3. Write $\partial \|\text{diag}(4, 0)\|_*$ and explain both orthogonality conditions on W .
4. Find $\mathcal{S}_2(\text{diag}(4, -2, 1))$. Which singular directions remain?
5. Why does the SVT proof use U_+, V_+ for the subgradient at the thresholded output, rather than all singular vectors of the input?
6. Derive the completion update for $t = 1/2$. What is the threshold, and what happens to an observed entry before SVT?
7. Explain why the proximal subproblem has a unique solution even when the full completion objective may have multiple minimizers.
8. In the example, how can a missing entry change when the data-fit gradient at that entry is zero?

4.15 Key formulas and further reading

4.15.1 The nuclear-norm foundation

For a compact SVD $X = U_r \Sigma_r V_r^\top$,

$$\|X\|_* = \sum_{i=1}^r \sigma_i(X) = \max_{\|G\|_{\text{op}} \leq 1} \langle G, X \rangle_F,$$

$$\partial \|X\|_* = \{U_r V_r^\top + W : U_r^\top W = 0, W V_r = 0, \|W\|_{\text{op}} \leq 1\}.$$

For a thin SVD $Y = U \text{diag}(\sigma_i) V^\top$,

$$\mathcal{S}_\tau(Y) = U \text{diag}((\sigma_i - \tau)_+) V^\top = \arg \min_Z \left\{ \frac{1}{2} \|Z - Y\|_F^2 + \tau \|Z\|_* \right\}.$$

4.15.2 The completion objective and its update

For

$$F(X) = \frac{1}{2} \|\mathcal{P}_\Omega(X - M)\|_F^2 + \lambda \|X\|_*, \quad \lambda > 0,$$

the smooth term has gradient $\mathcal{P}_\Omega(X - M)$ and Lipschitz constant $L = 1$. At step size $t = 1$, proximal gradient is

$$X^{(k+1)} = \mathcal{S}_\lambda \left(\mathcal{P}_\Omega(M) + \mathcal{P}_{\Omega^c}(X^{(k)}) \right).$$

At a general valid step $0 < t \leq 1$, the threshold is $t\lambda$. With exact updates and any minimizer X^* ,

$$F(X^{(k+1)}) \leq F(X^{(k)}) - \frac{1}{2t} \|X^{(k+1)} - X^{(k)}\|_F^2,$$

$$F(X^{(k)}) - F(X^*) \leq \frac{\|X^{(0)} - X^*\|_F^2}{2tk} \quad (k \geq 1).$$

The rate is an optimization guarantee for this objective. Identifying an unknown true matrix additionally depends on the observed entries, the underlying structure, and the choice of regularization.

4.15.3 Further reading

The definitions and derivation follow the supplied handwritten notes, with expanded proofs and a worked completion example. For standard treatments and algorithmic context:

1. J.-F. Cai, E. J. Candès, and Z. Shen, *A Singular Value Thresholding Algorithm for Matrix Completion*, SIAM Journal on Optimization, 20(4), 1956–1982, 2010. Author preprint. Section 2.1 establishes the nuclear-norm proximal formula.
2. R. Mazumder, T. Hastie, and R. Tibshirani, *Spectral Regularization Algorithms for Learning Large Incomplete Matrices*, Journal of Machine Learning Research, 11, 2287–2322, 2010. Journal article. This develops the Soft-Impute iteration used here.